Job Role
Embedded / IoT Design Engineer
Course Content from II-I to IV-I

Embedded / IoT Design Engineer

Course Content from II-I to IV-I

II-I (48 Hours) – Industrial Training Level #1

Foundation in Embedded Systems & IoT Programming

Model: Instructor-led 8 Days Training (48 Hours)
Methodology: Theory (30%) and Practice Labs (70%)

Pre-requisites

1. Basic programming fundamentals in any language (C/C++/Python/Java).
2. Basic understanding of digital electronics concepts.
3. Familiarity with PC usage, IDE installation, and serial communication tools.
4. Awareness of Wi-Fi and networking fundamentals.

Expected Outcomes

After completion of this semester, students will be able to:

1. Understand embedded systems architecture and applications.
2. Differentiate between microcontrollers and microprocessors.
3. Program ESP32 using Embedded C.
4. Configure GPIOs, PWM, Timers, and Interrupts.
5. Interface UART, I2C, and SPI peripherals.
6. Build Wi-Fi and Bluetooth enabled IoT applications.
7. Use cloud dashboards for data visualization.
8. Develop beginner-level IoT projects with real-time monitoring.

Day-wise Training Plan

Day 1 – Introduction to Embedded Systems & ESP32 Basics

1. Fundamentals of Embedded Systems
2. Embedded applications in Industry 4.0 and IoT
3. Microcontrollers vs Microprocessors
4. ESP32 Architecture Overview

Hands-On:

5. Installing Arduino IDE / ESP-IDF
6. ESP32 Board Configuration
7. LED Blink Program
8. Push Button Interfacing

Day 2 – Embedded C & GPIO Programming

1. Basics of Embedded C
2. GPIO Configuration and Registers
3. Pull-up / Pull-down Configuration
4. PWM Concepts and Applications

Hands-On:
1. LED Brightness Control using PWM
2. Interrupt-based Button Control

Day 3 – Timers & UART Communication
1. Timers and Counters in ESP32
2. Interrupt Handling
3. UART Communication Protocol
4. Motors and Drivers

Hands-On:
d. UART Communication with PC
e. GSM GPS Module Interfacing and Motors Interfacing

Day 4 – I2C & SPI Communication
1. I2C Protocol and Applications
2. SPI Protocol and Applications
3. Sensor Communication Concepts

Hands-On:
1. Temperature Sensor Interfacing using I2C
2. OLED Display Interfacing using SPI

Day 5 – Wi-Fi & Bluetooth in ESP32
1. TCP/IP Basics
2. Wi-Fi Modes in ESP32
3. Bluetooth Classic vs BLE 6.0
4. ESP32 in IoT Applications

Hands-On:
1. ESP32 Wi-Fi Connectivity
2. Bluetooth Data Transfer
3. Webserver-based LED Control

Day 6 – Cloud Connectivity & IoT Dashboards
1. Introduction to Cloud Platforms
2. MQTT Basics
3. IoT Dashboards and Monitoring

Hands-On:
1. Sending Sensor Data to ThingSpeak/Blynk
2. Dashboard Visualization
3. Real-time Data Monitoring

Day 7 – Sensor Integration & Data Logging

1. Analog and Digital Sensors
2. Data Acquisition Concepts
3. SD Card and EEPROM Basics
4. Introduction to Edge Data Processing

Hands-On:

1. DHT11/DHT22 Sensor Interfacing
2. Sensor Data Logging using ESP32
3. Threshold-based Alerts

Day 8 – Mini Project & Evaluation

1. IoT Node Design Flow
2. Testing and Debugging Basics
3. Introduction to Industrial IoT Applications

Mini Projects:

1. Smart Room Monitoring System
2. Wi-Fi-based Home Automation System
3. Bluetooth-controlled Device
4. Machine Health Monitoring
5. Sensor Data Logger with Cloud Dashboard

II-II (48 Hours) – Industrial Training Level #2

PCB Design & Microcontroller Interfacing

Model: Instructor-led 8 Days Training (48 Hours)
Methodology: Theory (45%) and Practice Labs (55%)

Pre-requisites

1. Completion of II-I Embedded Systems Training.
2. Understanding of GPIO, UART, I2C, and SPI concepts.
3. Basic knowledge of electronic components and circuits.
4. Familiarity with Embedded C programming.

Expected Outcomes

After completion of this semester, students will be able to:

1. Design PCB schematics and layouts using industry tools.
2. Apply PCB design rules and manufacturing constraints.
3. Generate Gerber files and BOM.
4. Understand EMI/EMC and thermal considerations.
5. Program Silicon Labs microcontrollers.
6. Interface sensors and peripherals with MCUs.
7. Build custom hardware boards for embedded applications.

8. Develop hardware-software integrated projects.

Day-wise Training Plan

Day 1 – PCB Fundamentals & Schematic Design

1. PCB Basics and Design Flow
2. Single Layer vs Multilayer PCB
3. Components, Footprints, Libraries
4. Overview of KiCad / Altium

Hands-On:

1. Schematic Design of LED Circuit
2. Footprint Assignment

Day 2 – PCB Layout & Design Rules

1. PCB Trace Routing Concepts
2. Ground Planes and Copper Pour
3. DRC and ERC Validation

Hands-On:

1. 5V Regulator Board Layout
2. Gerber File Generation

Day 3 – Advanced PCB Concepts

1. Signal Integrity Basics
2. EMI/EMC Concepts
3. Thermal Management
4. PCB Fabrication Workflow

Hands-On:

1. Sensor Interface PCB Design
2. PCB Validation Checks

Day 4 – Silicon Labs MCU Introduction

1. Overview of EFM32/EFR32 Controllers
2. Simplicity Studio IDE
3. GPIO and Clock Systems

Hands-On:

1. LED Blink using SiLabs MCU
2. Push Button Interfacing

Day 5 – Peripheral Interfacing

1. Timers and Interrupts
2. ADC/DAC Concepts
3. UART Communication

Hands-On:
1. Potentiometer Reading using ADC
2. UART Communication with PC

Day 6 – Sensor & Communication Interfaces
1. I2C and SPI Protocols
2. EEPROM and Sensor Communication
3. Low Power Features

Hands-On:
1. EEPROM Interfacing
2. SPI Sensor Integration

Day 7 – Power Electronics & Industrial Interfaces

1. Voltage Regulators and Power Supplies
2. Relay and Motor Driver Circuits
3. Industrial Sensors and Isolation Concepts
4. Introduction to CAN Protocol

Hands-On:

1. Relay Driver Circuit Design
2. Motor Driver PCB Module

Day 8 – Hardware Design Project & Evaluation

1. Hardware Testing and Debugging
2. PCB Assembly and Soldering Practices
3. Design Documentation Standards

Mini Projects:

1. Smart Energy Meter Interface Board
2. Sensor Acquisition PCB
3. Industrial Monitoring Interface Module

III-I (48 Hours) – Industrial Training Level #3

Advanced IoT Systems & Multilayer PCB Design

Model: Instructor-led 8 Days Training (48 Hours

Methodology: Theory (40%) and Practice Labs (60%)

Pre-requisites

1. Completion of II-I and II-II Training.
2. Familiarity with ESP32 and PCB Design.
3. Understanding of communication protocols.
4. Experience with sensor interfacing.

Expected Outcomes

After completion of this semester, students will be able to:

1. Design complete IoT system architectures.
2. Implement MQTT-based communication systems.
3. Configure secure IoT communication.
4. Build cloud-connected IoT applications.
5. Design multilayer PCBs.
6. Apply EMI/EMC and power integrity concepts.
7. Integrate multiple IoT nodes.
8. Develop industry-oriented IoT prototypes.

Day-wise Training Plan

Day 1 – IoT System Architecture & MQTT Basics

1. IoT Architecture Layers
2. MQTT Protocol and QoS
3. Device-Gateway-Cloud Architecture

Hands-On:

1. Installing MQTT Broker
2. ESP32 MQTT Publish/Subscribe

Day 2 – IoT Communication & Security

1. HTTP, CoAP, WebSocket Protocols
2. TLS/SSL Security
3. Authentication and Encryption

Hands-On:

1. Secure MQTT Communication
2. IoT Dashboard Creation

Day 3 – Edge Computing & Sensor Networks

1. Edge Processing Concepts
2. Multi-node IoT Systems
3. Industrial IoT Use Cases

Hands-On:

1. Multi-sensor ESP32 Node
2. Node-RED Visualization

Day 4 – Multilayer PCB Design

1. 2-layer vs 4-layer PCB
2. Stack-up Design
3. High-Speed Routing

Hands-On:

1. 4-layer PCB Layout Design

Day 5 – EMI/EMC & Thermal Design

1. EMI Sources and Mitigation
2. Thermal Management
3. Power Integrity Concepts

Hands-On:

1. High-current PCB Design
2. Decoupling and Filtering

Day 6 – Cloud IoT Platforms & Analytics

1. AWS IoT / Azure IoT Overview
2. Data Logging and Analytics
3. Real-time Monitoring Systems

Hands-On:

1. Cloud Dashboard Integration
2. Sensor Data Analytics

Day 7 – Industrial IoT & Automation

1. Industrial Communication Protocols
2. Smart Factory Concepts
3. Predictive Maintenance Basics
4. IoT in Smart Agriculture and Healthcare

Hands-On:

1. IoT-based Predictive Monitoring Demo
2. Industrial Sensor Network Setup

Day 8 – IoT Hardware Integration Project

1. Complete IoT Product Development Cycle
2. PCB Validation and Testing
3. Product Deployment Concepts

Mini Projects:

1. Smart Agriculture Monitoring System
2. Industrial Equipment Monitoring System
3. Smart Energy Management System

III-II (48 Hours) – Industrial Training Level #4

Advanced Microcontrollers & Real-Time Embedded Applications

Model: Instructor-led 8 Days Training (48 Hours)
Methodology: Theory (40%) and Practice Labs (60%)

Pre-requisites

1. Strong Embedded C fundamentals.
2. Understanding of MCU peripherals and communication protocols.
3. Experience with ESP32 and Silicon Labs MCUs.
4. Basic understanding of IoT systems.

Expected Outcomes

After completion of this semester, students will be able to:

1. Work with TI and STM32 microcontrollers.
2. Configure ADC, PWM, DMA, Timers, and Interrupts.
3. Interface communication peripherals.
4. Develop low-power embedded applications.
5. Build STM32-based sensor nodes.
6. Apply debugging and optimization techniques.
7. Understand DMA and advanced peripheral management.
8. Prepare for RTOS and advanced firmware development.

Day-wise Training Plan

Day 1 – Introduction to TI Microcontrollers

1. MSP430 and Tiva Architecture
2. GPIO and Clock Systems
3. Code Composer Studio

Hands-On:

1. LED Blink and GPIO Interfacing

Day 2 – Peripheral Interfacing & Low Power Modes

1. ADC and PWM Modules
2. Watchdog Timers
3. Interrupt Handling
4. Low Power Modes

Hands-On:

1. Analog Sensor Reading
2. PWM-based Motor Control

Day 3 – Communication Protocols

1. UART Communication
2. SPI Communication
3. SD Card Interfacing

Hands-On:

1. UART-based Data Transfer
2. SPI SD Card Logging

Day 4 – STM32 Architecture & Development

1. ARM Cortex-M Architecture
2. STM32CubeIDE and HAL Libraries
3. Clock and GPIO Configuration

Hands-On:

1. STM32 LED and Push Button Control

Day 5 – STM32 Peripheral Interfacing

1. ADC and PWM Configuration
2. Timers and Interrupts
3. DMA Basics

Hands-On:

1. Temperature Sensor Interfacing
2. PWM LED Control

Day 6 – Advanced Communication & Debugging

1. UART Register-level Understanding
2. Debugging Techniques using ST-Link

3. Serial Monitoring and Logging

Hands-On:

1. STM32 to PC Communication
2. Debugging Peripheral Failures

Day 7 – Firmware Optimization & RTOS Readiness

1. Embedded Firmware Architecture
2. Memory Optimization Techniques
3. Introduction to RTOS Concepts
4. Task Scheduling Basics

Hands-On:

1. Cooperative Task Scheduling Demo
2. Performance Measurement

Day 8 – Embedded Application Project & Evaluation

1. Embedded Product Testing Flow
2. Firmware Validation and Documentation
3. Industrial Coding Standards

Mini Projects:

1. Smart Sensor Node using STM32
2. Data Acquisition System
3. Motor Speed Control System

IV-I (48 Hours) – Industrial Training Level #5

RTOS, Embedded Linux & Industrial IoT Systems

Model: Instructor-led 8 Days Training (48 Hours)
Methodology: Theory (40%) and Practice Labs (60%)

Pre-requisites

1. Strong knowledge of STM32 programming.
2. Understanding of DMA, UART, I2C, SPI, and Interrupts.
3. Familiarity with embedded networking concepts.
4. Exposure to IoT systems and cloud connectivity.

Expected Outcomes

After completion of this semester, students will be able to:

1. Develop RTOS-based embedded applications.

2. Configure multitasking systems using FreeRTOS.
3. Implement task synchronization mechanisms.
4. Work with Raspberry Pi and Embedded Linux.
5. Interface Linux-based systems with sensors and peripherals.
6. Develop end-to-end IoT gateway applications.
7. Build scalable embedded and industrial IoT systems.
8. Become industry-ready for Embedded Firmware and IoT roles.

Day-wise Training Plan

Day 1 – Advanced STM32 Concepts

1. NVIC and SysTick
2. Interrupt Priority Handling
3. DMA Architecture

Hands-On:

1. DMA-based ADC Transfer
2. UART Communication between STM32 Boards

Day 2 – SPI & I2C Communication

1. SPI Protocol Deep Dive
2. I2C Addressing and Communication

Hands-On:

1. EEPROM Interfacing
2. SPI Sensor Communication

Day 3 – RTOS Fundamentals

1. Task Scheduling
2. Context Switching
3. FreeRTOS Architecture

Hands-On:

1. FreeRTOS Setup on STM32
2. Multi-task LED Control

Day 4 – Task Communication & Synchronization

1. Queues and Semaphores
2. Mutexes and Event Groups
3. ISR-to-Task Communication

Hands-On:

1. Sensor and Display Task Communication

Day 5 – Embedded Linux & Raspberry Pi

1. Linux Architecture
2. Embedded Linux Concepts
3. Raspberry Pi Setup and Configuration

Hands-On:

1. GPIO Programming using Python
2. Push Button and LED Interface

Day 6 – IoT Gateway & Cloud Connectivity

1. MQTT and HTTP Communication
2. Raspberry Pi as IoT Gateway
3. Cloud Data Logging

Hands-On:

1. Sensor Data Upload to Cloud
2. Dashboard Visualization

Day 7 – Beaglebone Black & Embedded Linux Development

1. Boot Process: ROM → U-Boot → Kernel
2. Device Tree Concepts
3. Buildroot and Cross Compilation

Hands-On:

1. GPIO and Peripheral Interfacing
2. LCD Interfacing on Beaglebone

Day 8 – Capstone Project & Industry Evaluation

1. Embedded Product Development Lifecycle
2. Industrial Documentation and Deployment
3. Performance Optimization and Validation

Capstone Projects:

1. Industrial IoT Monitoring System
2. RTOS-based Smart Monitoring Device
3. IoT Gateway using STM32 + Raspberry Pi
4. Predictive Maintenance System using MQTT Dashboard

IV-I (72 hours) – Placement Readiness Program

1. Company Specific Practice Tests – 48 h

- Technical aptitude – 16 h
- Domain-specific assessments – 16 h
- Coding and problem-solving for Embedded roles – 16 h

2. Soft Skills – 24 h

- Resume building and LinkedIn optimization – 4 h
- Group discussions and technical presentations – 8 h
- Mock interviews (technical + HR rounds) – 12 h

The course is designed by Dr. Mukesh Sukla, Director, VLSIPro Technologies Pvt. Ltd. and Former Senior Manager, Marquee Semiconductor